Experiment 11. Pandas use cases

In [1]:
```
import numpy as np
import pandas as pd
list=[[1,'Smith',50000],[2,'Jones',60000]]
df=pd.DataFrame(list)
df
df.columns=['Empd','Name','Salary']
df
df.info()
df=pd.read_csv("/content/50_Startups.csv")
df.info()
df.head()
df.tail()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   Empd    2 non-null      int64 
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64 
dtypes: int64(2), object(1)
memory usage: 180.0+ bytes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     object 
 4   Profit           50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

Out[1]:

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	45173.06	California	14681.40

In [2]:
```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/employee.csv")
df.head()
```

```
df.tail()
df.info()
df.salary
type(df.salary)
df.salary.mean()
df.salary.median()
df.salary.mode()
df.salary.var()
df.salary.std()
df.describe()
df.describe(include='all')
empCol=df.columns
empCol
emparray=df.values
emparray
employee_DF=pd.DataFrame(emparray,columns=empCol)
employee_DF
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype 
---  ------  --------------  ----- 
 0   emp id  7 non-null      int64 
 1   name    7 non-null      object
 2   salary  7 non-null      int64 
dtypes: int64(2), object(1)
memory usage: 300.0+ bytes
```

Out[2]:

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000